Project – Manufacturing Assembly Line

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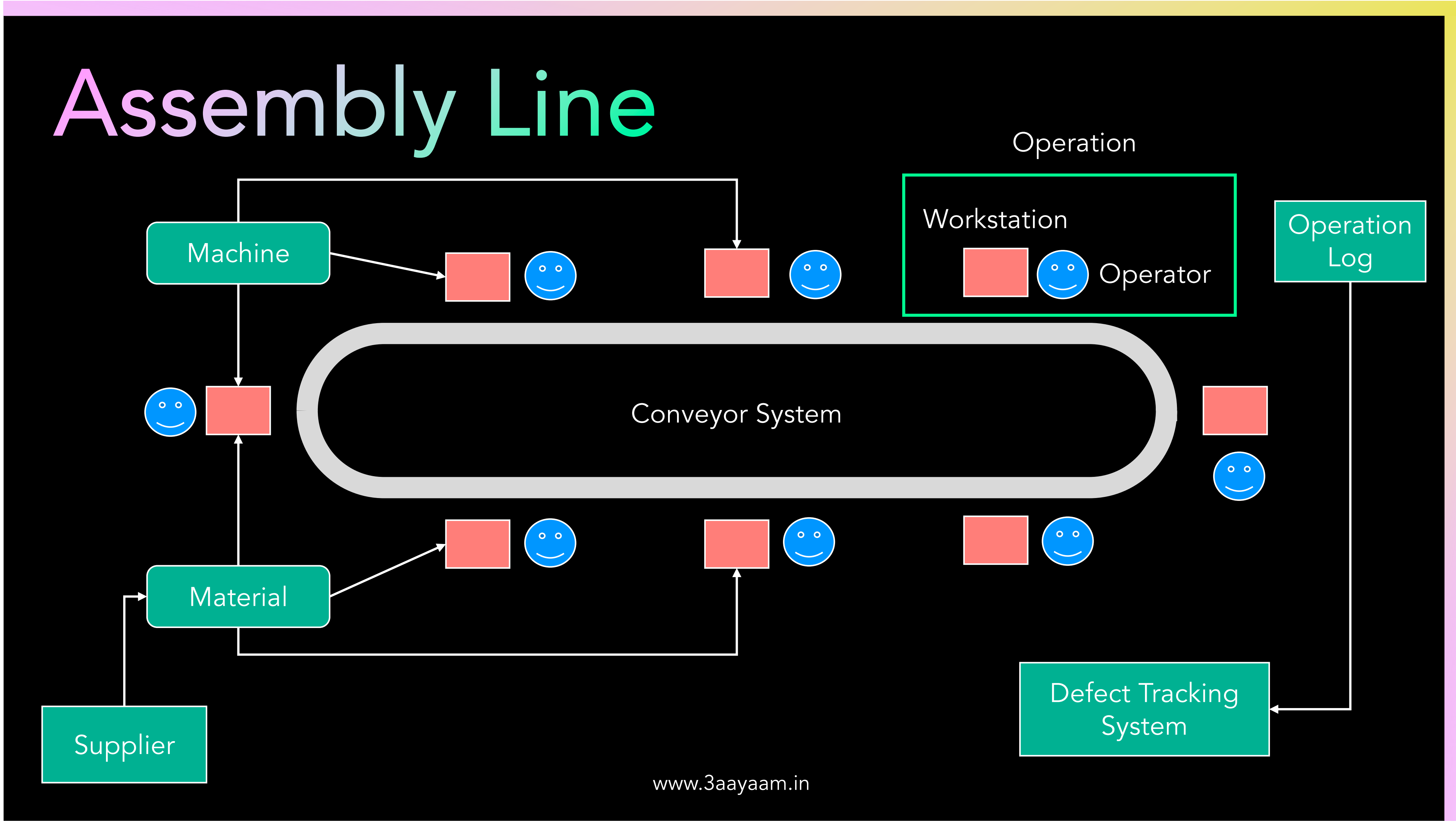
Assembly Line
Operation
Workstation
Operator
Operation Log
Machine
Conveyor System
Material
Supplier
Defect Tracking System
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Schema

Supplier

```
{
  "supplier_id" : <...>,
  "supplier_machine_id" : <...>,
  "supplier_details" : [...]
}
```

Material

material_id
supplier_id
quantity
unit_cost
delivery_date
current_stock
min_stock_level
reorder_point

Operation Log

```
{
  "oplog_id": <...>,
  "work_order_id": <...>,
  "machine_id": <...>,
  "work_order_details":
     {
       "operator_id": <...>,
       "oplog_start_time": <...>,
       "oplog_end_time": <...>,
       "output_qty": <...>,
       "defect_count": <...>,
       "defect_percent": <...>
     }
}
```

Operator

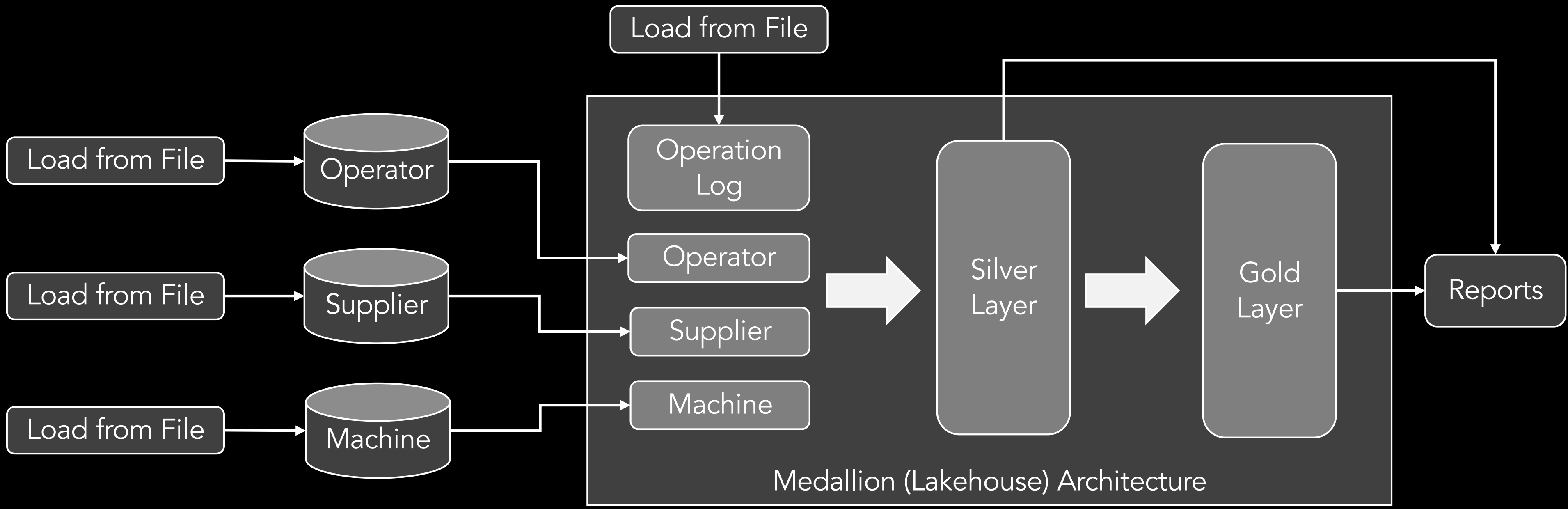
operator_id
operator_skill_level
operator_shift

Machine

machine_id
machine_type
machine_status
last_maintenance_date

Architecture

- Cannot use Data Warehouse for cost concerns. Want to implement Lakehouse Architecture.



Use Case

- Perform initial load using data_1 set. Use data_2 set load new records with the following considerations:
  - New records should be inserted in DB and Lakehouse.
  - Existing records, if changed, should be appended in both DB and Lakehouse.
  - Existing records, if not changed, should be ignored.
- Management needs a clear, real-time view of the production line to identify bottlenecks and optimize efficiency. They need to know which machines or operators are causing slowdowns as they happen.
- Management wants to see the alignment of operator skill level and time taken to finish a particular work and check if there is any corelation.
- Generate pro-active order details based on the usage on assembly line. Output quantity in operation log can be thought of as material qty.